

Émilienne Al Wazzan

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EDUCATION

University of Cambridge | *MEng Computer Science* 2019-2023

- Developed distributed systems capstone project handling 50K+ concurrent requests with 99.9% uptime using gRPC and Kubernetes
- Led systems programming society with 120+ members, organizing 8 technical workshops on API design patterns

Imperial College London | *BSc Computer Science (Exchange Year)* 2021-2022

- Completed advanced algorithms and cloud computing specialization with distinction
- Contributed to 3 peer-reviewed papers on microservices architecture optimization

EXPERIENCE

Stripe | *Senior API Engineer* 2023-Present

- Architected and deployed billing API v3 handling \$2.8B in monthly transaction volume across 180+ countries, reducing latency by 38% through query optimization and caching strategies
- Led cross-functional team of 4 engineers to implement rate limiting system that decreased API abuse incidents by 92% while maintaining 99.95% availability for legitimate requests
- Refactored legacy webhook delivery system, increasing reliability from 97.2% to 99.98% and reducing failure notifications by 65% through exponential backoff and circuit breaker patterns

Cloudflare | *API Engineer II* 2021-2023

- Built RESTful API gateway processing 45TB of DNS and security logs daily, optimizing database queries to reduce response times by 44% and server costs by \$220K annually
- Designed comprehensive API documentation and SDK generation framework adopted by 150+ enterprise customers, reducing integration time by 73% and support tickets by 58%
- Implemented distributed tracing and monitoring for 12 microservices using OpenTelemetry, enabling 21% faster incident resolution and 84% reduction in API error rates

HashiCorp | *API Engineer* 2020-2021

- Developed gRPC-based configuration API supporting 40+ enterprise clients with 99.99% SLA, enabling infrastructure-as-code workflows for 500+ daily active users
- Implemented automated API versioning and backward compatibility framework reducing schema migration issues by 91% and enabling 2-week release cycles instead of monthly
- Created comprehensive integration test suite covering 280+ API endpoints, increasing code coverage from 52% to 87% and catching 34 critical bugs before production

PROJECTS

AsyncIO Event Router | *Python, FastAPI, Redis, PostgreSQL, Docker* 2023-Present

- Open-source event streaming router handling 100K events/second with <50ms latency, achieving 2.3K GitHub stars and adopted by 18 production companies
- Reduced memory footprint by 56% through intelligent batching and implemented persistent retry logic decreasing message loss from 0.3% to 0.001%

GraphQL-to-REST API Transpiler | *TypeScript, Node.js, GraphQL, Jest, AWS Lambda* 2022-2023

- Built open-source tool enabling automatic conversion of GraphQL queries to optimized REST endpoints, reducing API calls by 67% for 25+ enterprise adopters
- Published 3 technical articles reaching 12K developers, establishing thought leadership and driving 340+ GitHub contributions from community

Rate Limiter as a Service | *Go, Redis Cluster, Protocol Buffers, gRPC, Prometheus* 2021-2022

- Engineered distributed rate limiting microservice with <1ms p99 latency across multi-region deployments, handling 2.1M requests/second with zero data loss
- Reduced infrastructure costs by 48% through token bucket optimization and increased customer adoption by 156% through simplified SDK interface

API Gateway Performance Analyzer | *Rust, Tokio, InfluxDB, Grafana, WASM* 2020-2021

- Developed real-time API performance profiler analyzing 500K+ requests daily, identifying bottlenecks and recommending optimizations saving customers 31% in latency
- Achieved 89% adoption rate among Stripe partners through automated insights and reduced customer support burden by 42 weekly hours

EXTRACURRICULAR

- API Design Conference (APIDays) | Conference Speaker & Workshop Facilitator2022-Present
- Delivered keynote on 'Scaling APIs to Billions of Requests' reaching 2,300+ attendees and generating 145K video views, establishing expertise in high-throughput systems
 - Facilitated 6 workshops on gRPC, REST best practices, and API security with 18% attendee satisfaction rating and sponsorship partnerships worth €85K
- London DevOps & Cloud Native Community | Technical Lead & Open Source Mentor2020-Present
- Mentored 12 junior engineers contributing to distributed systems projects, resulting in 8 promotions and 3 successful open-source project launches
 - Organized monthly meetups attracting 200+ engineers, curated technical content on microservices and Kubernetes, increasing community membership by 340%

TECHNICAL SKILLS

Languages – Python, Go, TypeScript/Node.js, Java, Rust, SQL, Bash

Technologies – REST APIs, gRPC, GraphQL, Kafka, Redis, PostgreSQL, MongoDB, Elasticsearch, Protocol Buffers, OAuth 2.0, JWT, Docker, Kubernetes, AWS (Lambda, RDS, EC2), GCP, Microservices Architecture

Tools – Git, GitHub, GitLab, JIRA, Datadog, New Relic, Prometheus, Grafana, OpenTelemetry, Postman, Kong, Swagger/OpenAPI, Apache Airflow, Jenkins, GitHub Actions